Accuracy of a requirements ambiguity detector: case of a pet social network

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Abstract

Context: Ambiguity is one of the most persistent defects in natural-language requirements specifications, and rule-based detectors are frequently proposed as a low-cost mechanism to flag it before requirements reach design. Evidence on how closely such detectors reproduce human expert judgement remains scarce for Spanish-language specifications produced in Latin American settings.
Objective: This study measures the extent to which a rule-based ambiguity detector replicates the consensus judgement of independent human experts over the complete requirements corpus of MundiPets, a real social network for responsible pet adoption and breeding, and characterises the patterns of disagreement between the two.
Methods: We conducted a paired quasi-experiment over 50 requirements (25 functional, 16 non-functional and 9 design constraints) specified according to an eight-attribute template. A purpose-built rule-based detector, implemented in Python and frozen before data collection, classified each requirement as ambiguous or non-ambiguous using four lexical and syntactic rules. In parallel, three independent software engineers, blind to the detector output and to one another, classified the same requirements in randomised order using a shared rubric. The protocol was preregistered on the Open Science Framework before any data were

collected. Agreement was quantified with precision, recall, $\boldsymbol{F_1}$, Cohen's $\boldsymbol{\kappa}$ and Fleiss' $\boldsymbol{\kappa}$.
Results: *[Deferred to Deliverable 4 (2B): quantitative results are reported in version 2.0 of this manuscript, once the full statistical analysis described in Section 3.7 has been executed through the versioned scripts.]*
Conclusion: *[Deferred to Deliverable 4 (2B): conclusions follow from the results.]*

Keywords: Requirements Engineering, Empirical Software Engineering, Natural Language Processing, Interrater Agreement, Requirements Quality

1 Introduction

MundiPets is a social network for responsible pet adoption and breeding, currently under specification for veterinary practices and pet owners in the province of Los Ríos, Ecuador. The system lets owners publish verified pet profiles, lets registered veterinary practices validate the associated medical records, and supports adoption and breeding requests through a staged workflow that includes compatibility assessment by an artificial-intelligence component. Requirements engineering matters acutely in this domain because a defect in a requirement about medical-record validation or about the visibility of an owner's location does not merely degrade usability: it propagates into a system that handles animal health data and personal data protected under Ecuador's Organic Law on Personal Data Protection [**?**]. Specifying such a system with verifiable, unambiguous requirements is therefore a precondition for lawful operation rather than a stylistic preference.

Ambiguity remains one of the defects most consistently reported in natural-language requirements. The systematic mapping study of Montgomery et al. [**?**] shows that, although requirements quality has been studied extensively, empirical work concentrates on a small number of quality attributes and rarely validates automated detection against human judgement on complete, real specifications. Atoum et al. [**?**] reach a convergent conclusion in their systematic review of requirements quality assurance, identifying the scarcity of validated automated support as an open challenge. Ferrari et al. [**?**] demonstrated that ambiguity in elicitation is entangled with tacit knowledge, which suggests that purely lexical detection has an intrinsic ceiling; yet the size of that ceiling has not been quantified for specifications written in Spanish, nor for corpora produced by novice analysts, the setting in which a large share of requirements documents are in fact written. This is the gap the present study addresses.

This paper makes three contributions, each grounded in evidence collected by the authors and released as an open dataset:

1. a quantified comparison between a rule-based ambiguity detector and the blind consensus of three independent human experts over a complete requirements corpus of 50 items — functional requirements, non-functional requirements and design constraints — written in Spanish;

2. a characterisation of the disagreement patterns between automated and human classification, reported per requirement and per rule, which exposes the specific linguistic phenomena that a rule-based approach systematically over- or under-flags;
3. a replication package — corpus, detector output, expert classifications, analysis scripts and preregistered protocol — released under CC BY 4.0 so that the study can be repeated on other corpora and other languages.

The study answers two research questions:

RQ1 With what precision, recall and F_1 does a rule-based automatic detector replicate the consensus judgement of human experts when classifying the requirements of MundiPets as ambiguous?

RQ2 Which disagreement patterns predominate between the automatic detector and expert judgement, and what do they reveal about the limits of a rule-based approach relative to human judgement in identifying ambiguity?

The remainder of the paper is organised as follows. Section 2 positions the study within the closest related work. Section 3 describes the experimental design, participants, instruments and analysis plan. Section 4 reports the results, Section 5 interprets them, Section 6 discusses threats to validity, and Section 7 concludes.

2 Related work

This section is not a full systematic literature review; it follows the abbreviated protocol of Kitchenham and Charters [**?**], reporting a documented search string, explicit inclusion and exclusion criteria, the number of studies retrieved and screened, and a comparative table of the closest works.

2.1 Search strategy

The search string combined three concept blocks:

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("requirements engineering" OR "software requirements") AND ("ambiguity" OR
"requirements smell" OR "requirements quality") AND ("automatic detection"
OR "rule-based" OR "natural language processing")
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The string was executed on Scopus and Google Scholar on 2 August 2026, restricted to the 2023–2026 period. Inclusion criteria: peer-reviewed studies reporting empirical data on the quality, ambiguity or automated analysis of natural-language requirements, published in requirements-engineering or software-engineering venues. Exclusion criteria: editorials, short contributions under four pages, and studies without empirical data. The search retrieved 689 records in Scopus and 3,670 in Google Scholar. After title and abstract screening, 30 records were read in full text and 6 were retained. These were combined with eight studies already identified in the project bibliography, yielding the 14 entries compared in Table 1.

2.2 Comparison with the closest studies

Table 1 contrasts the retained studies with the present work. The first eight rows correspond to studies already identified in the project bibliography; the last six were retained from the search described above.

The niche of the present study is narrow and, to our knowledge, unoccupied: a paired comparison between a frozen, fully documented rule-based detector and the blind consensus of independent experts, over the *complete* requirements corpus of one real system, specified in Spanish, with the corpus, the detector output and the individual expert classifications released as open data. Existing work either maps the field without primary agreement data, or evaluates detectors on English-language benchmark corpora that were not produced by the team reporting the evaluation.

3 Methodology

The empirical component follows Approach 2 of the course framework — automatic detection of ambiguity and smells in requirements — and is designed as a paired quasi-experiment, reported following the guidelines of Jedlitschka et al. [**?**] and the experimental design principles of Wohlin et al. [**?**]. The protocol was preregistered on the Open Science Framework [**?**] on 27 July 2026 at 10:49:41 (Ecuador time), before the classification rubric was distributed to the expert panel and before the detector was executed over the corpus.

3.1 Research questions in PICOC format

3.2 Hypotheses

RQ1 is tested through the following pair of hypotheses:

H_0 The classification produced by the automatic ambiguity detector is not associated with the consensus classification of the experts; the observed agreement between the two (Cohen's κ) is not significantly different from chance agreement ($\kappa = 0$).
H_1 The classification produced by the automatic detector is significantly associated with the expert consensus classification; the observed agreement is significantly greater than chance ($\kappa > 0$).

RQ2 is exploratory and is not subjected to hypothesis testing; it is answered through qualitative analysis of the cases where detector and consensus diverge.

3.3 Variables

The *independent variable* is the classification method applied to each requirement, with two levels: (a) rule-based automatic detector and (b) consensus of the three human experts. The *dependent variable* is the resulting binary classification of each requirement as ambiguous or non-ambiguous, from which the agreement metrics are computed.

Four *control variables* were fixed. First, requirement type (FR, NFR, DC) was recorded to verify that detector performance does not vary systematically with the artefact evaluated. Second, the detector configuration — the four rules listed in Section 3.4 — remained invariant throughout execution; no rule was adjusted after observing partial results. Third, the presentation order of the requirements was randomised independently for each rater to control fatigue and order effects. Fourth, raters were blind both to the detector output and to one another's classifications until their own was complete.

3.4 Design and materials

The corpus comprises the complete set of 50 requirements of the MundiPets specification: 25 functional requirements, 16 non-functional requirements and 9 design constraints. Each requirement was assigned an anonymous identifier (REQ-01 to REQ-50), stripped of its type label, and presented to the raters in randomised order. A key linking anonymous identifiers to the real requirement identifiers and to the detector output was retained by the research team and not shared with the raters.

The detector is a purpose-built Python program based on regular expressions. It flags a requirement as ambiguous when at least one of four rules fires: (i) presence of a vague quantifier; (ii) multiple conjunctions concealing more than one requirement in a single statement; (iii) passive voice or absent subject; and (iv) absence of an explicit verification criterion. No large language model was used at any point of the classification pipeline, which keeps every classification decision traceable to a named rule.

3.5 Participants and recruitment

The expert panel comprised three software engineers holding a professional degree from the Universidad Técnica Estatal de Quevedo (two women and one man), none of whom belonged to the MundiPets team or participated in writing the requirements under evaluation. Inclusion criteria were: (a) a professional degree in software engineering or a closely related field; (b) demonstrable training or experience in requirements specification and analysis; and (c) availability to complete the classification within the study schedule.

Each rater signed, on 28 July 2026, an informed consent form specific to this role, drafted in compliance with Ecuador's Organic Law on Personal Data Protection [**?**], explicitly authorising the anonymised use of their classifications in peer-reviewed publications and their deposit in an open dataset.

3.6 Instruments

Two instruments were used. The *expert classification rubric* is a workbook containing an instruction sheet, a reference-criteria sheet for identifying ambiguity, and an evaluation sheet where each rater records, per requirement, the identifier, the full text, the classification, the type of smell detected, a brief justification and a self-reported confidence level on a 1–5 scale. The *expert consent template* adapts the consent form used in the previous elicitation round to the rater role, adding the clause on anonymised use in peer-reviewed publications.

3.7 Statistical analysis plan

Because the central variable is categorical, the primary analysis is based on agreement metrics rather than mean comparison.

Primary analysis (RQ1, H_0/H_1). Precision, recall and F_1 of the detector are computed against expert consensus, defined by majority vote across the three raters. The hypothesis is tested through a significance test of Cohen's κ (statistic $z = \kappa/SE_\kappa$, against the standard normal distribution). The magnitude of agreement is interpreted as the effect size using the Landis and Koch scale: 0.00–0.20 slight; 0.21–0.40 fair; 0.41–0.60 moderate; 0.61–0.80 substantial; 0.81–1.00 almost perfect.

Inter-rater agreement. Cohen's κ [**?**] is computed for each pair of raters and Fleiss' κ [**?**] across the three, interpreted on the same scale. Low expert agreement would relativise any conclusion about detector performance, since the reference consensus itself would be in doubt.

Normality test. The Shapiro–Wilk test is applied to the continuous confidence variable (1–5 scale) recorded in the rubric, as a precondition for the secondary analysis.

Secondary analysis (RQ2, exploratory). To characterise disagreement, the confidence level reported by the raters is compared between requirements where the detector agrees with consensus and those where it diverges. If Shapiro–Wilk does not reject normality, an independent-samples t test is used with Cohen's d as effect size; otherwise, the Mann–Whitney U test with Cliff's δ. This is complemented by a qualitative note on each disagreeing requirement.

Correction for multiple comparisons. If agreement metrics are additionally reported disaggregated by requirement type (FR, NFR, DC), the resulting p values from those three related comparisons are adjusted with the Benjamini–Hochberg procedure.

Power. Following Molléri et al. [**?**] and Wohlin et al. [**?**], $\alpha = 0.05$ and $1 - \beta = 0.80$ are fixed. Because the population size is bounded by the real specification corpus rather than by an expandable recruitment target, power is reported in reverse: instead of fixing a target N, we determine the minimum detectable κ that the design can identify at that α and power, given the available $N = 50$.

Reproducibility. All computations are executed through versioned analysis scripts; no figure or table is produced manually.

4 Results

[Deferred to Deliverable 4 (2B): this section reports the analysed data without interpretation, organised by research question, with every table and figure generated by the versioned analysis scripts. It is written only after the full analysis plan of Section 3.7 has been executed.]

5 Discussion

[Deferred to Deliverable 4 (2B): implications for research, implications for practice, and surprising or counter-intuitive findings. Every claim must be traceable to a specific datum in Section 4 or to a cited reference.]

6 Threats to validity

*[Deferred to Deliverable 4 (2B): internal, external, construct and conclusion validity following Wohlin et al. [**?**], with the mitigation applied and the residual limitation acknowledged for each threat. The anticipated threats are already documented in the preregistered protocol and are carried over here once the results are known.]*

7 Conclusions and future work

[Deferred to Deliverable 4 (2B): one paragraph per research question, one consolidating the contributions promised in Section 1, and one with at least three concrete lines of future work.]

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Declarations

- **Funding.** This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.
- **Competing interests.** The authors declare no competing interests.
- **Ethics approval and consent to participate.** The study was conducted under the ethics framework of the Requirements Engineering course at the Universidad Técnica Estatal de Quevedo. All participants signed an informed consent form drafted in compliance with Ecuador's Organic Law on Personal Data Protection [**?**].
- **Consent for publication.** All participants explicitly authorised the use of their anonymised data in peer-reviewed publications and its deposit in an open dataset.
- **Data availability.** The preregistered protocol is available on the Open Science Framework at https://osf.io/khyf2/overview The anonymised dataset — labelled requirements corpus, detector output, individual expert classifications, traceability matrix and analysis scripts — will be deposited on Zenodo under a CC BY 4.0 licence with an assigned DOI upon acceptance, following the FAIR principles [**?**] and the software citation principles of Smith et al. [**?**].
- **Materials availability.** Not applicable.
- **Code availability.** The detector and the analysis scripts are available in the project repository at https://github.com/andymendozap03/MundiPets-PFC-IngReq

- **Author contribution.** E.D.F.A.: conceptualization, investigation, writing — original draft. G.A.G.O.: software (design and implementation of the rule-based detector). A.J.M.P.: methodology, visualization (system modelling and diagrams). G.A.M.S.: investigation, data curation (interviews and transcriptions). J.S.N.S.: formal analysis, data curation, project administration (protocol preregistration and statistical analysis). G.C.G.U.: supervision, validation. All authors reviewed and approved the final manuscript.

Table 1 Comparison with the closest related studies

Reference	Year	Study type	Population	Difference with our work
Montgomery et al. [?]	2022	Systematic mapping	Primary studies on requirements quality	Maps the field but reports no primary comparison between an automated detector and expert consensus on a single complete corpus.
Atoum et al. [?]	2021	Systematic review	Literature on requirements quality assurance and validation	Identifies the lack of validated automation as an open challenge; does not measure detector–expert agreement empirically.
Ferrari et al. [?]	2016	Empirical study	Elicitation interviews	Studies ambiguity as it arises in elicitation and its link to tacit knowledge; our unit of analysis is the written requirement in a closed specification.
Fischbach et al. [?]	2023	Tool evaluation	Conditional statements in natural-language requirements	Derives acceptance tests automatically; shares the rule/NLP approach but targets test generation, not ambiguity classification.
Gobov [?]	2023	Practitioner survey	Specification and modelling techniques in practice	Characterises which techniques practitioners use; provides no automated quality measurement.
Said et al. [?]	2026	Model evaluation	SRS documents	Classifies requirements as functional or non-functional with a learned model; our detector targets ambiguity and is rule-based and fully auditable.
Cheng et al. [?]	2026	Systematic review	Generative AI applied to requirements engineering	Surveys generative approaches; we deliberately avoid large language models to keep every classification traceable to a named rule.
Vogelsang and Fischbach [?]	2024	Methodological guideline	LLM use for NLP tasks in RE	Proposes usage guidelines rather than reporting an agreement study.
Abdeahad et al. [?]	2026	Model evaluation	7,061 requirements from the Fault-Prone SRS benchmark	Achieves high accuracy with a deep model, but evaluates against dataset labels rather than against a blind expert panel, and the model is not auditable rule by rule.
Fantechi et al. [?]	2023	Tool comparison	Three example requirements documents	Closest to our design: compares a rule-based tool with an LLM on ambiguity. Uses the authors themselves as the reference judgement, whereas our reference is a blind panel independent of the specification team.
Talha et al. [?]	2025	Tool evaluation	1,553 requirements across 17 domains	Targets anaphoric and coordination ambiguity with a transformer; our detector targets lexical and syntactic smells and is compared against human consensus rather than a labelled benchmark.
Alemneh and Berhanu [?]	2024	Systematic review	Literature on requirements smells and detection techniques	Catalogues detection techniques; reports no primary agreement measurement on a real corpus.
Intana et al. [?]	2024	Tool evaluation	Thai-language SRS documents	The closest non-English precedent; confirms that language-specific evaluation is needed but does not cover Spanish nor report inter-rater agreement.
Bashir et al. [?]	2025	Industrial study	Three railway requirements datasets	Reports the over-flagging pattern (high recall, low precision) also observed in rule-based tools; uses LLMs, which we deliberately avoid for auditability.

Table 2 Research questions structured with the PICOC framework

Element	Description
Population	The set of functional requirements (FR), non-functional requirements (NFR) and design constraints (DC) specified for the MundiPets system, written according to the eight-attribute template of the course and current at the time the experiment was executed.
Intervention	Automatic classification of each requirement as ambiguous or non-ambiguous by a rule-based detector that identifies vague quantifiers, multiple conjunctions concealing more than one requirement, passive voice, and absence of an explicit verification criterion.
Comparison	Manual and independent classification of the same requirement set by three experts in requirements engineering, with no prior access to the detector output (blind evaluation).
Outcome	Precision, recall and F_1 of the automatic detector against expert consensus; inter-rater agreement via Cohen's κ between pairs and Fleiss' κ across the three raters.
Context	Academic project of the Requirements Engineering course (Universidad Técnica Estatal de Quevedo, 2026–2027 academic period) applied to the real MundiPets system, with the specification written in Spanish.